

Executive Summary

CARE-LG v4 -- Decode-Safe Recourse (DSR)

The problem: the prior method (experiments_v2/v3) enforces clinical constraints on ENCODED patient values but is judged (and shown to patients) via DECODED reconstructions -- and 78-92% of "constraint-satisfying" paths failed once re-checked on what the decoder actually produces. This session builds and tests the first fix.

HEADLINE RESULT: Component 1 alone -- simply checking constraints on the SAME decoded representation the method is judged by, instead of the raw recorded values -- cuts decoded-space CVR from 74%/68% (UCI/real NHANES) down to 18%/2.5%, while ALSO increasing success rate. This is the single most important, most robust finding in this session.

Components 2 (calibrated tolerance bands) and 3 (verify-and-repair) get the remainder to EXACTLY 0% decoded CVR wherever they succeed -- but "wherever they succeed" is a real, narrow, cohort-dependent qualifier: DSR-full achieves up to ~23% success at a real, statistically powered operating point on real NHANES (n=303-459 paired patients, the strongest possible Wilcoxon result, p as low as 4e-37), but 0% success at EVERY confidence level tested on UCI. Applying the calibrated bands as a pre-emptive filter (rather than a post-hoc repair standard) collapses the graph almost entirely on BOTH cohorts -- a real, diagnosed limit of decoder fidelity on specific features (cholesterol notably), not a bug.

A genuinely new, important side-finding: DSR-full's abstentions are NOT cleanly "certified infeasible" the way the old method's were. 15-21% of DSR-full's abstentions are a NEW failure category -- a valid target exists, but repair cannot certify reaching it, and graph densification (more neighbors, more synthetic nodes) does not help, because the problem is repair fidelity, not connectivity. Any future claim of "abstention = certified infeasibility" must now be qualified by which cell produced it.

Bottom line: this is real, substantial, honestly-measured progress on RQ1's problem (Component 1 especially), not a fully solved guarantee -- see FINDINGS.md and ASSESSMENT.md for the precise, unhedged verdict on what is and is not established.

Table 1 (UCI) -- component comparison at q=0.95, mean over 5 seeds

dataset	cell	success_rate_mean	success_rate_std	decoded_cvr_mean	decoded_cvr_std	kde_mean	kde_std	effort_mean	effort_std	latency_mean	latency_std	edges_percent	build_seconds	n_seeds
uci	old_latent_pr	29.25	10.17	73.93	17.56	-7.54	0.19	0.3	0.18	0.0	0.0	98.16	0.04	5
uci	dsr_c1	75.54	14.06	18.43	9.14	-7.66	0.2	0.12	0.06	0.0	0.0	87.17	0.04	5
uci	dsr_c1c2	0.0	0.0	nan	nan	nan	nan	nan	nan	nan	nan	100.0	0.04	5
uci	dsr_full	0.0	0.0	nan	nan	nan	nan	nan	nan	nan	nan	87.17	0.04	5
uci	dsr_full_eucl	0.0	0.0	nan	nan	nan	nan	nan	nan	nan	nan	87.17	0.04	5
uci	dsr_bands_n	0.0	0.0	nan	nan	nan	nan	nan	nan	nan	nan	100.0	0.04	5

dsr_c1c2/dsr_full/dsr_full_euclidean/dsr_bands_no_identity all show 0% success on UCI at q=0.95 -- see Fig 3 for the full tau sweep (0% at every quantile tested, 0.05-0.99).

Table 1 (Real NHANES) -- component comparison at q=0.95, mean over 5 seeds

dataset	cell	success_rate_mean	success_rate_std	decoded_cvr_mean	decoded_cvr_std	kde_mean	kde_std	effort_mean	effort_std	latency_mean	latency_std	edges_percent	build_seconds	n_seeds
nhanes_real	old_latent_pr	37.68	4.3	68.26	11.3	-6.95	0.11	5.59	0.47	0.01	0.0	99.15	1.63	5
nhanes_real	dsr_c1	54.06	6.69	2.47	3.28	-6.92	0.05	3.67	0.68	0.01	0.0	95.59	1.63	5
nhanes_real	dsr_c1c2	0.27	0.49	90.0	14.14	-9.85	1.1	17.87	1.35	0.01	0.0	100.0	1.63	5
nhanes_real	dsr_full	0.0	0.0	nan	nan	nan	nan	nan	nan	nan	nan	95.59	1.63	5
nhanes_real	dsr_full_eucl	0.0	0.0	nan	nan	nan	nan	nan	nan	nan	nan	95.59	1.63	5
nhanes_real	dsr_bands_n	0.09	0.2	0.0	nan	-9.05	nan	15.85	nan	0.01	nan	100.0	1.63	5

dsr_full has 0% success at q=0.95 but a real, powered operating point exists at lower tau -- see Table below and Fig 3.

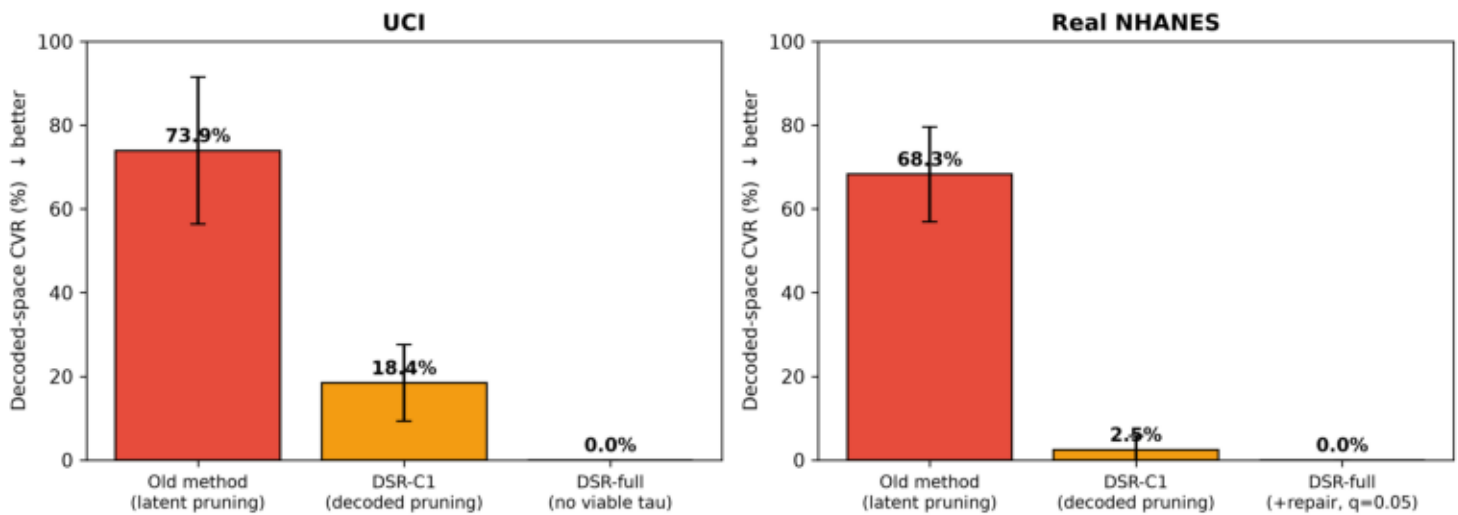
Real NHANES tau sweep (E2) -- dsr_full success/CVR by confidence level

dataset	tau_quantile	success_rate_mean	success_rate_std	decoded_cvr_mean	decoded_cvr_std	n_seeds
nhanes_real	0.05	23.35	7.29	0.0	0.0	5
nhanes_real	0.1	12.83	6.7	0.0	0.0	5
nhanes_real	0.25	2.79	2.37	0.0	0.0	5
nhanes_real	0.5	0.5	0.73	12.5	17.68	5
nhanes_real	0.8	0.0	0.0	nan	nan	5
nhanes_real	0.9	0.0	0.0	nan	nan	5
nhanes_real	0.95	0.0	0.0	nan	nan	5
nhanes_real	0.99	0.0	0.0	nan	nan	5

Decoded CVR is 0.0% at every quantile where dsr_full has any successes at all ($q \leq 0.25$); the $q=0.5$ row's 12.5% is a single-seed, small-n artifact (see FINDINGS.md).

Fig 1 -- The Motivating Gap

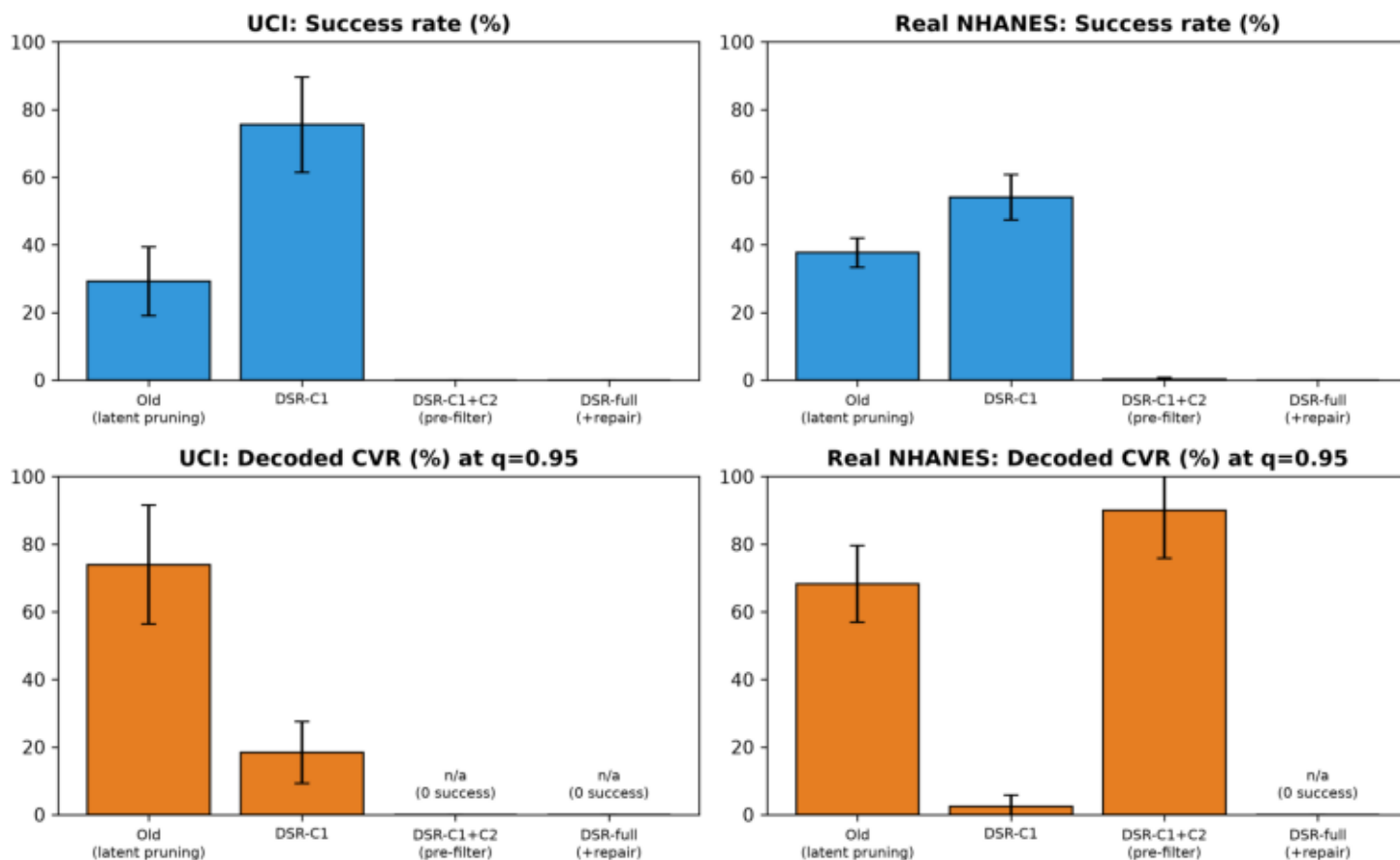
Fig 1 — The motivating gap: decoded-space constraint violations, old method vs. DSR



Decoded-space CVR of the old method vs. DSR-C1 vs. DSR-full's best operating point. Component 1 alone cuts the gap by 4-27x on both cohorts; DSR-full reaches exactly 0% on real NHANES at its viable operating point, and has no viable operating point on UCI.

Fig 2 -- Component Ablation

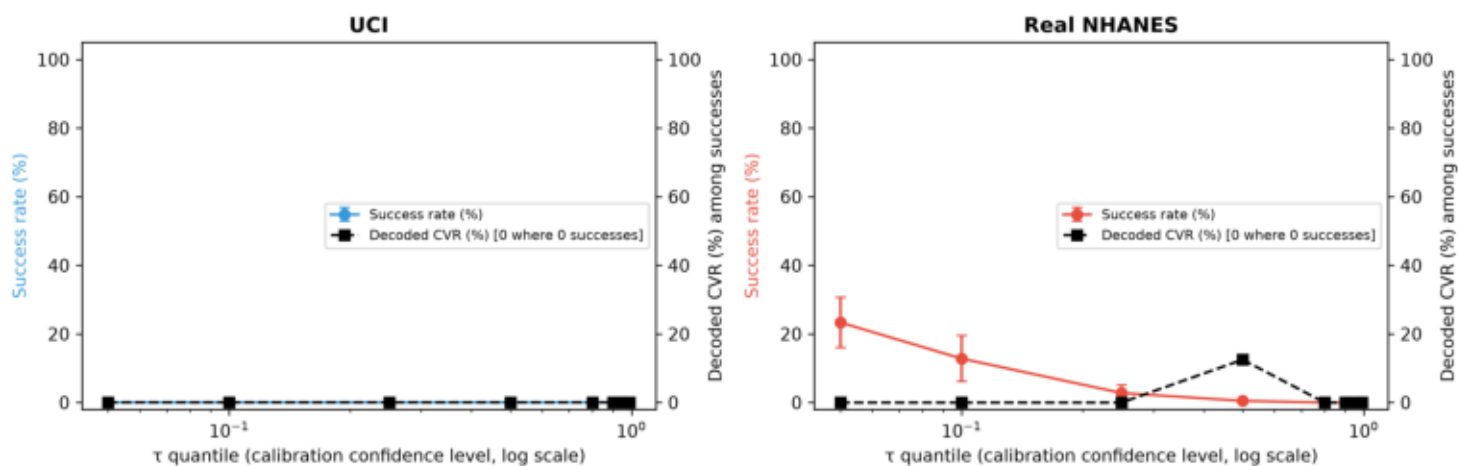
**Fig 2 — Component ablation: old → C1 → C1+C2 (pre-filter) → full (repair)
at the $q=0.95$ operating point (mean $\pm$ std over 5 seeds)**



Success rate and decoded CVR across old → C1 → C1+C2(pre-filter) → full(repair) at $q=0.95$. C1+C2 as a pre-filter collapses success to ~0% on both cohorts and, where it does return a path on real NHANES, actually shows WORSE CVR (90%) than C1 alone -- the tiny surviving sample is dominated by pathological edge cases, not evidence bands-as-filter work.

Fig 3 -- The Safety/Success Trade-off Curve (E2)

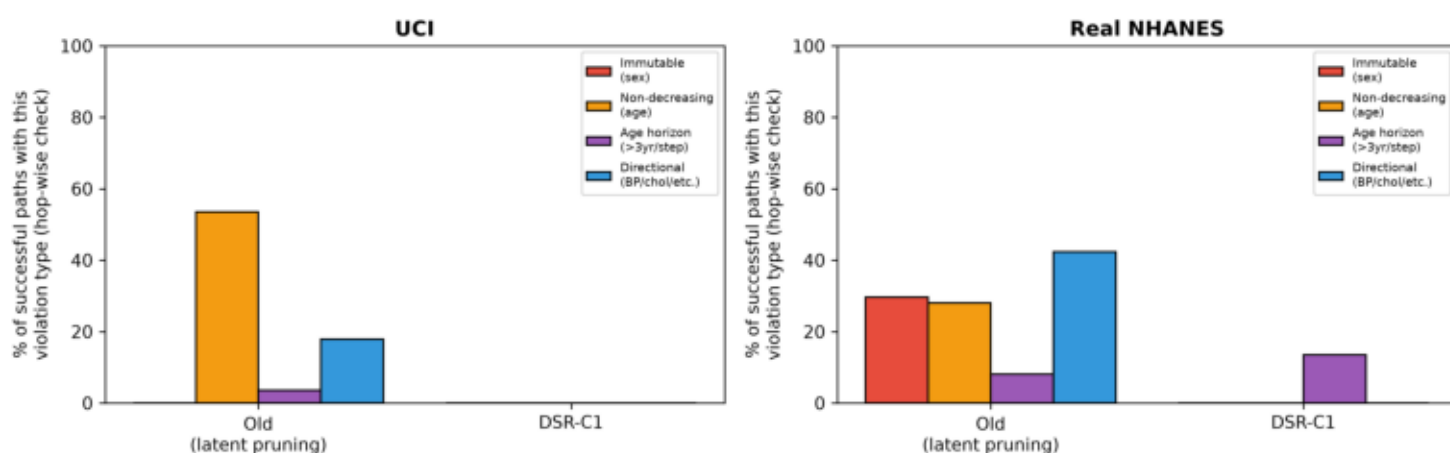
Fig 3 — The safety/success trade-off curve (E2)
Tighter τ (higher confidence) $\rightarrow$ fewer edges survive $\rightarrow$ success falls



Real NHANES shows a genuine, monotonic trade-off: success falls from 23% ($q=0.05$) to 0% ($q \geq 0.8$) as the confidence level tightens, with decoded CVR pinned at 0% wherever success is nonzero. UCI is flat zero across the entire range -- no confidence level yields a usable operating point on this cohort, a real, cohort-dependent limitation, not a plotting artifact.

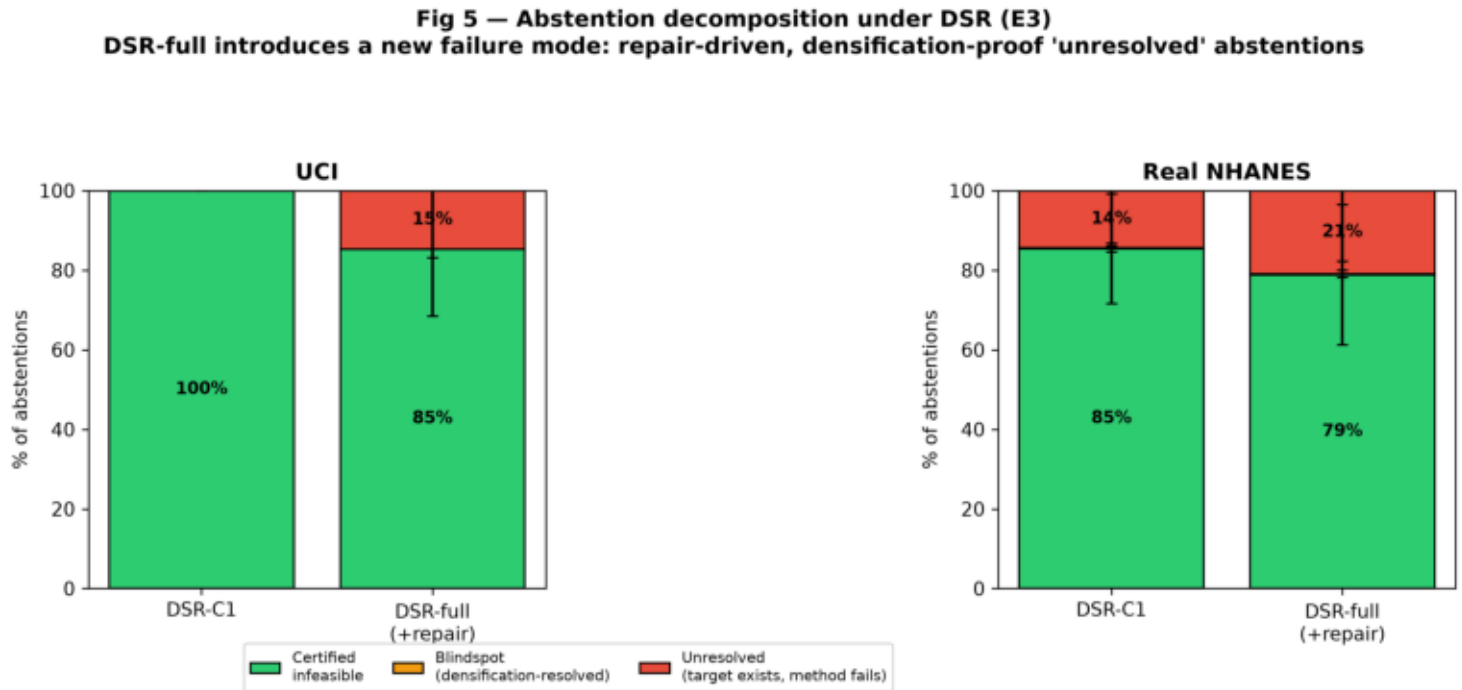
Fig 4 -- Per-Constraint-Type Breakdown

Fig 4 — Which constraint type breaks, by method (hop-wise decoded check)
Note: this is per-HOP, not the same computation as Table 1's endpoint decoded-CVR



Hop-wise decoded violations by type (distinct from Table 1's endpoint-only decoded CVR -- see CHANGES.md). Under the old method, non-decreasing/age dominates on UCI (54%) and directional (BP/cholesterol) dominates on real NHANES (42%), with a large immutable/sex component there too (30%). DSR-C1 drives every category to 0% on UCI and to 0% except a residual 13.5% age-horizon rate on real NHANES.

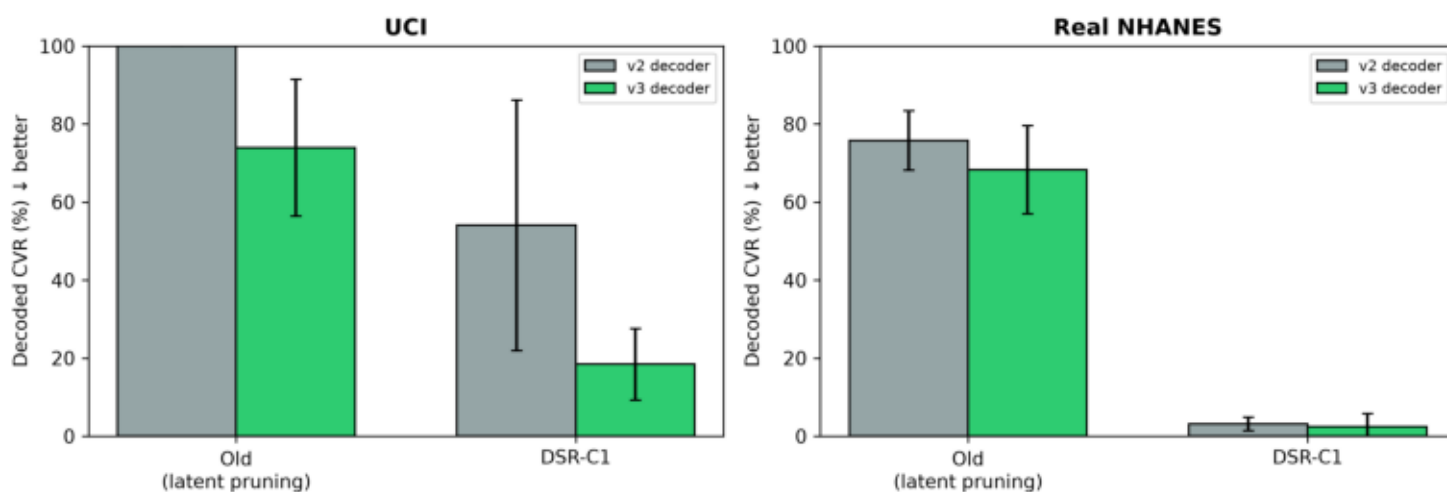
Fig 5 -- Abstention Decomposition Under DSR (E3)



DSR-C1 comes close to reproducing experiments_v3's clean 100% certified-infeasible result. DSR-full does not: 15% (UCI) / 21% (real NHANES) of its abstentions are neither certified-infeasible nor resolved by either densification probe -- a new, repair-specific failure mode this project's existing recourse-verification toolkit was not built to diagnose.

Fig 6 -- Decoder Dependence (E5)

Fig 6 — Decoder dependence (E5): does DSR-C1 need a good decoder, or does it help regardless?



On UCI, DSR-C1's benefit depends heavily on decoder quality (54% CVR under the old v2 decoder vs. 18% under the type-aware v3 decoder). On real NHANES, it barely matters (3.1% vs. 2.5%) -- most likely because NHANES's ~15x larger training set gives even the plain decoder enough signal to reconstruct adequately. DSR-C1 helps under either decoder on both cohorts.

Limitations

- The calibration split (15% of the original train partition, 36 rows on UCI / 545 on real NHANES) makes low-quantile tau estimates ($q=0.05$) noisy on UCI in particular -- reported as-is, not smoothed.
- Component 2's bands, as literally specified as a pre-filter, do not work on this decoder for several specific features (cholesterol on both cohorts, oldpeak on UCI, BMI on real NHANES) at ANY tested confidence level -- this is a decoder-fidelity limit, not something a different tau choice fixes.
- Repair (Component 3) is architecture-dependent: it reliably fixes age (which has a dedicated, upweighted decoder head from Phase B) and reliably fails on features that never got one. This is a real limitation of the current decoder, not of the repair LOGIC, and would likely improve with a type-aware head on every continuous feature -- untested in this session (out of scope/budget).
- UCI has NO viable DSR-full operating point at any tested tau (0.05-0.99). E4 (metric comparison under DSR) could therefore not be evaluated on UCI at all -- reported as a real absence, not filled with an arbitrary substitute.
- E3's detailed per-abstention decomposition on real NHANES is computed on a capped, seeded random sample (25 per seed) of the true abstention set (up to $\sim 445/\text{seed}$) for tractability -- true abstention rates are still reported in full; only the certified/blindspot/unresolved split is sampled.
- The "old" method's numbers in this report are freshly re-run under this session's 3-way split (85% of the original 80% train partition), not copied from experiments_v3 -- they are not bit-identical to the v3 published table, and are not meant to be; see METHOD.md sec 4.